TEXT ANALYTICS (CISB5123)

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ASSIGNMENT 1: PROJECT 1

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1.0 Problem Identification

The increasing use of Google Maps as a platform for sharing dining experiences has generated a large volume of unstructured textual reviews. In the Malaysian food and beverage industry, these reviews provide important information about customer opinions on food quality, service performance, cleanliness, atmosphere, and pricing. Although such feedback is valuable, the large number of reviews makes manual analysis difficult, slow and prone to inconsistency. In this project, more than 222,021 reviews are available, creating a clear information overload problem in which meaningful insights are hidden within a massive collection of raw text.

This issue is significant because customer reviews can strongly influence restaurant reputation and consumer decisions. Businesses that fail to understand recurring complaints or positive feedback may lose opportunities to improve service quality and customer satisfaction. Manual review analysis is not practical at this scale, as it may lead to missed patterns, subjective interpretation, and human error. Therefore, text analytics provides a suitable approach to process and organize the data systematically. By applying text preprocessing and analytical techniques, this project aims to convert unstructured food review data into useful insights that can support better decision-making in the restaurant industry.

1.1 Project Objectives

This project focuses on using text analytics to process and analyze restaurant reviews from Google Maps. Since the reviews are written in free-text form, they often contain noisy and unstructured data that cannot be analyzed directly. Therefore, the reviews must first be cleaned and transformed into a structured format before sentiment analysis and topic modeling can be carried out. The results can help restaurant owners better understand customer opinions and improve their services based on real feedback:

- To develop a robust text preprocessing pipeline to convert "noisy" Malaysian restaurant reviews into a structured format for analysis.
- To identify prevailing customer sentiment (Positive/Negative) across various restaurant locations to benchmark service quality.
- To extract key themes (Topic Modeling) such as service, pricing, and taste to provide data-driven recommendations for restaurant owners.

1.2 Project Drivers

1.2.1 Market Competition

The Malaysian food and beverage industry is highly competitive, especially in areas with a high concentration of restaurants such as Ipoh. Customers are often presented with many dining

options, so restaurants must continuously improve their food quality, service, and overall customer experience to remain competitive. Online reviews provide direct feedback from customers, making them an important source of information for identifying strengths and weaknesses. This strongly supports the project objective of analyzing restaurant reviews to uncover useful insights that can help businesses improve and differentiate themselves.

1.2.2 Social Media Influence and Online Reputation

Platforms such as Google Maps play an important role in shaping consumer decisions because customers frequently rely on online ratings and reviews before choosing a restaurant. Positive reviews can attract more customers, while negative reviews can damage a restaurant's reputation and reduce trust. As a result, understanding customer sentiment and common discussion themes in online reviews has become a critical business need. This is highly relevant to the project objective, as the analysis aims to identify sentiment patterns and extract major themes from review data to better understand customer perceptions.

1.2.3 Operational Efficiency and Scalability

The dataset used in this project contains more than 222,021 reviews, making manual analysis unrealistic, time-consuming, and inconsistent. Restaurant owners or analysts would need significant time and effort to read, categorize, and summarize such a large volume of text. By applying text preprocessing and automated text analytics techniques, the project enables faster and more scalable analysis of customer feedback. This directly supports the project objective of building a structured approach to transform raw review data into meaningful information for decision-making.

1.3 Project Scope

1.3.1 Problem Addressed

This project focuses on addressing the challenge of extracting meaningful insights from a large volume of unstructured restaurant reviews collected from Google Reviews. The main issue is that customer feedback is written in free-text form, making it difficult to manually identify overall sentiment, repeated complaints, and common discussion themes across many restaurants and locations. The project is limited to analyzing textual review data and does not cover financial performance, restaurant operations, or non-textual customer behavior.

1.3.2 Insights to Extract

The project aims to extract three main types of insights from the review data. First, it will identify customer sentiment, such as positive and negative opinions, to understand overall satisfaction levels. Second, it will detect common keywords and recurring themes related to issues such as food quality, service, pricing, cleanliness, and dining experience. Third, it will compare review patterns across different restaurant locations to observe differences in customer perception. The project focuses on text-based insights only and does not attempt to predict sales, customer retention, or future business outcomes.

1.3.3 Data Sources and Target Audience

The dataset is sourced from publicly available Google Reviews related to restaurants. It contains customer-written feedback, ratings, restaurant names, and locations, which provide a relevant basis for text analytics. The primary target audience for this project includes restaurant owners, restaurant managers, and hospitality industry consultants who need a systematic way to understand customer opinions and improve service quality. The project is intended to support decision-making based on customer feedback rather than replace direct business evaluation methods.

1.3.4 Project Boundaries

This project is limited to the preprocessing and analysis of restaurant review text within the selected dataset. It focuses only on the available review records and does not include real-time data collection, multilingual translation, fake review detection, advanced predictive modeling, or direct recommendation system development. The study is also limited to insights that can be derived from textual content and associated metadata such as ratings and locations.

1.3.4 Ethical Considerations

Several ethical considerations are acknowledged in this project. Although the reviews are publicly accessible, the analysis focuses on aggregated patterns rather than individual reviewers in order to respect user privacy. Reviewer identities are not the subject of the study, and personal details are not emphasized in reporting. In addition, the dataset may contain bias because it only reflects the opinions of users who chose to post online reviews, which may not represent all customers. There may also be subjective, exaggerated, or inconsistent opinions in the reviews, which can affect the fairness and accuracy of the findings.

2.0 Data Collection Instrument

The dataset used in this project was obtained from Kaggle, which provides Google Review about Malaysian restaurant.

The dataset includes textual reviews along with rating information provided by customers.

Attribute	Description
Author	Name of reviewer
Rating	Rating given by user
Review	Textual customer feedback
Restaurant	Restaurant name
Location	Restaurant location

The dataset is stored in CSV format, which allows for easy processing using Python.

2.1 Sampling

The dataset contains 222,021 customer reviews, which provides a large sample size suitable for text analytics research.

All records in the dataset were used for analysis to ensure comprehensive coverage of customer feedback and opinions.

Using the entire dataset helps capture diverse perspectives and improves the reliability of the analysis results.

2.2 Dataset Description

The dataset consists of customer reviews that describe user experiences at different restaurants. Each record includes five attributes: author name, rating, review text, restaurant name, and location. These attributes provide both supporting context and direct customer feedback, allowing the dataset to be used for restaurant-related text analysis.

The Review column is the most important attribute in this project because it contains unstructured text written by customers. This text reflects customer satisfaction or dissatisfaction with different aspects of the restaurant experience, such as food, service, and environment. Since unstructured text cannot be analyzed directly, it must first go through preprocessing before text analytics techniques can be applied.

2.3 Text Preprocessing

Text preprocessing is required to transform raw textual data into a structured format suitable for analysis.

Several preprocessing steps were applied:

Text Cleaning
- Punctuation marks, special characters, and unnecessary symbols were removed. All text was converted to lowercase to ensure consistency.

Tokenization
- Review sentences were split into individual words (tokens) for easier processing.

Stop Word Removal
- Common words such as "the", "is", and "and" were removed because they do not add significant meaning to the analysis.

Stemming
Words were reduced to their root form to standardize similar words.

Example:
"liked", "liking", "likes" → "like"

These preprocessing steps help improve the quality of text data and prepare it for further analysis.